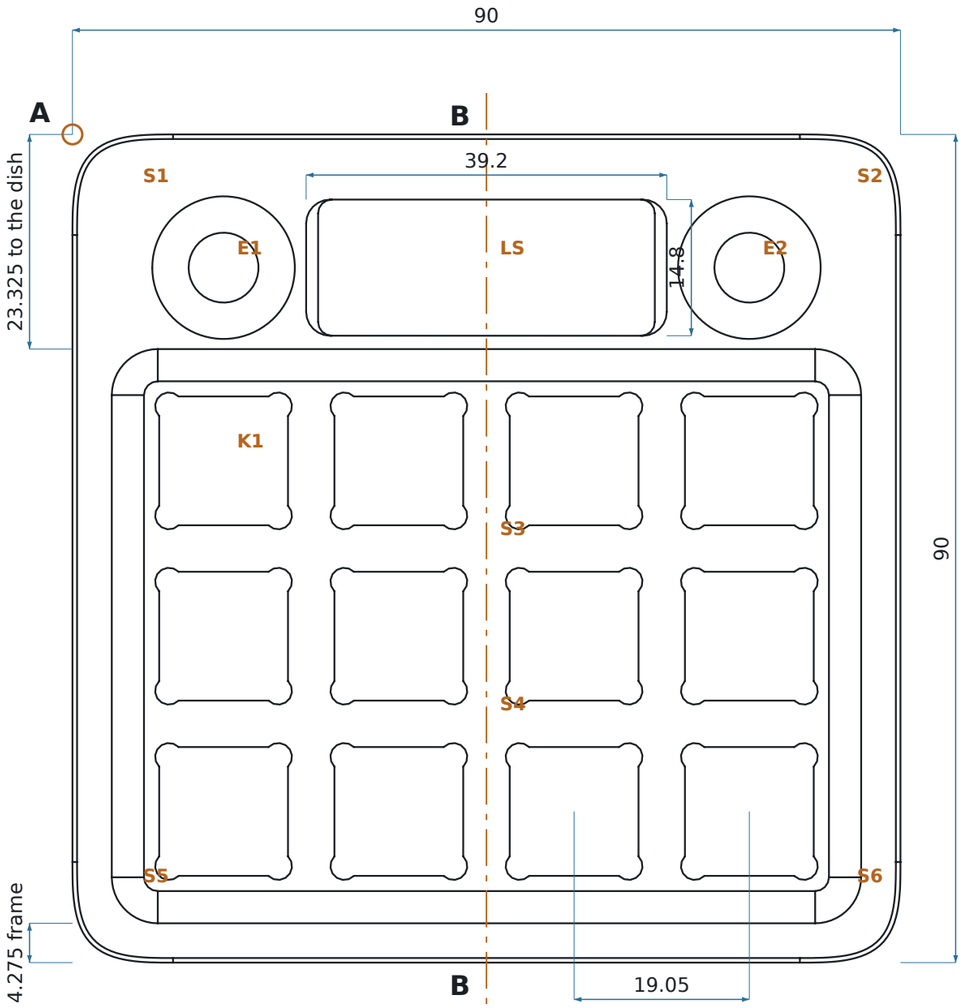
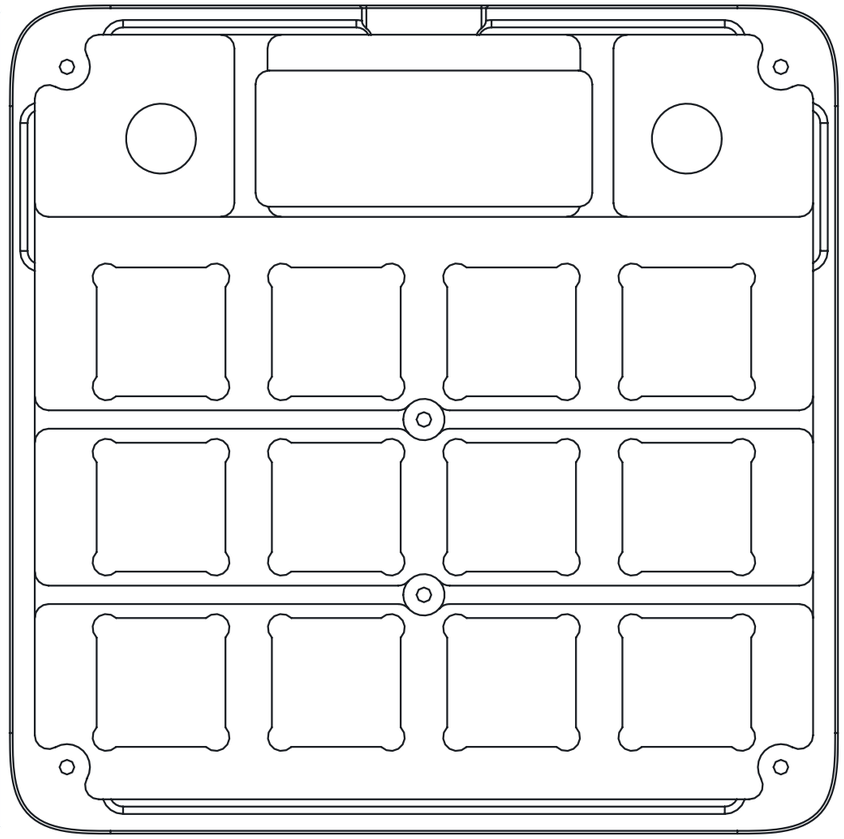
FROM ABOVE



FROM BELOW (mirrored left to right)

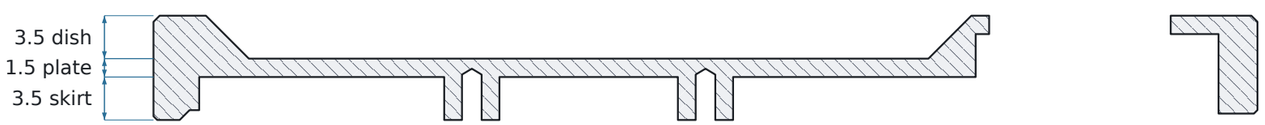


macropad case: top, CNC variant

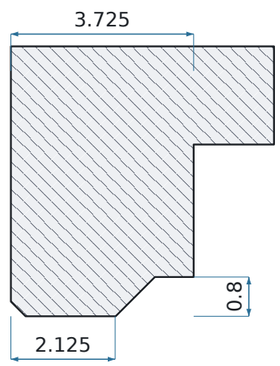
Holes and recesses, from datum A

tag	X	Y
ø1.6 x 3.7 from below (point to 4.18), tap M2 to 3 full thread		
S1	6.2	6.675
S2	83.8	6.675
S3	45	45
S4	45	64.05
S5	6.2	82.775
S6	83.8	82.775
ø7.6 through; well ø15.5 x 0.7 deep from above		
E1	16.425	14.475
E2	73.575	14.475
14 square through, first of 12 on a 19.05 grid		
K1	16.425	35.475
lens recess 39.2 x 14.8, R2.6, 1.25 deep from above		
opening 36.6 x 14.8 through, in its middle		
LS	45	14.475

SECTION B-B



GAP AND LAP JOINT, front wall, 8:1



Scale 1.5:1 plans, 2:1 section, 8:1 detail, on A3. Dimensions in mm, computed from the model.
Datum A: back-left corner of the box round the outline, seen from above. X to the right, Y toward the front.
3-axis, two setups: underside up, then top side up.
Every pocket corner is R1.5 or larger; the switch cut-outs have corner reliefs of that radius.
Switch cut-outs 14 square in a 1.5 mm plate. Take their tolerance from the Cherry MX datasheet.
Key dish: flat floor on the plate, walls at 45 degrees, 3.5 deep. A 90 degree chamfer mill of 8 mm or more cuts the walls.
Screw holes: drill 1.6 to 3.7 at full diameter, tap M2 with a bottoming tap to 3 full thread or more, for M2 x 8.
Setup 2 (top side up): hold the part on a fixture plate through the six M2 holes, or in soft jaws cut to the outline. The skirts are too thin for vise pressure.
Outline corners: one quintic Bezier each, control points 10.94, 6.56 and 2.19 mm from the box corner on both sides. Take the curve from the STEP file.
General tolerance: ISO 2768-m is a proposal. Nothing has been made or measured.

MATERIAL AND FINISH (a proposal; nothing has been made)
6061-T6 aluminium.
Fine bead blast, 120 to 150 grit glass bead, then Type II sulphuric anodise, 10 to 15 um (a guess), dyed clear (Silver) or grey (Space grey).
Anodise grows each face by about 0.01 to 0.02 mm (a guess). Every fit on this sheet leaves 0.1 mm or more per side, so no fit is cut to allow for it.
Break unmarked edges 0.05. The 0.3 chamfers are cosmetic: measure them after blasting.
Mask the six tapped M2 holes before anodising, or chase them with an M2 6H tap after it.
LENS OPENING
Cut it in setup 2, after the lens recess, with the same 3 mm end mill, down to Z 6.35 (2.15 deep). The recess and the opening then share their long walls without a step between setups.